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# Assessing Mobile Application Importance for Future Flooding Using Statistical Analysis and GIS: A study in Fort Plain, New York

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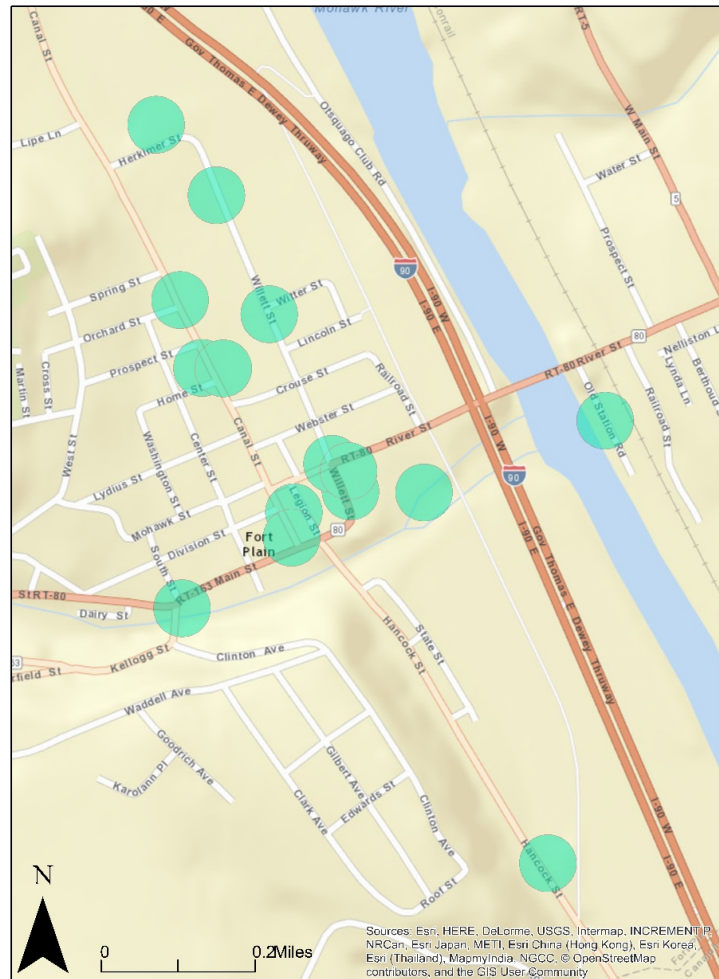
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## Introduction

On June 28, 2013, there was an occurrence of heavy rainfall across the Mohawk River Valley, where it fell at nearly one inch per hour (Weather.gov); at times the rain varied between three to five inches. The result was heavy flooding across the Mohawk Valley and the Adirondacks. As a result, parts of the New York State Thruway between exits 29A and 29 were closed. In addition to this, many of the roads were washed out. The entire village of Fort Plain went under water (Weather.gov). As a result of the flooding, 87 businesses were washed out, and were not expected to reopen, and over a hundred homes suffered major damage (Allen, 2013; Weather.com, 2013). Fort Plain is a village within Montgomery County, New York and is located near the Mohawk River. When living near flood hazards, evacuation routes are crucial for emergency preparedness. The purpose of this study is to assess the importance of evacuation routes, given a mobile application. This mobile application will be also based on a simulation and reconstruction of flood damage in a local area, which is a similar technique previously used in this area (D'Orsa et al., 2016). In addition to this, the mobile application will guide Fort Plain workers to highly elevated evacuation shelters through accessible roads at the time of a flood hazard. A statistical analysis of a survey has shown the importance of this application.

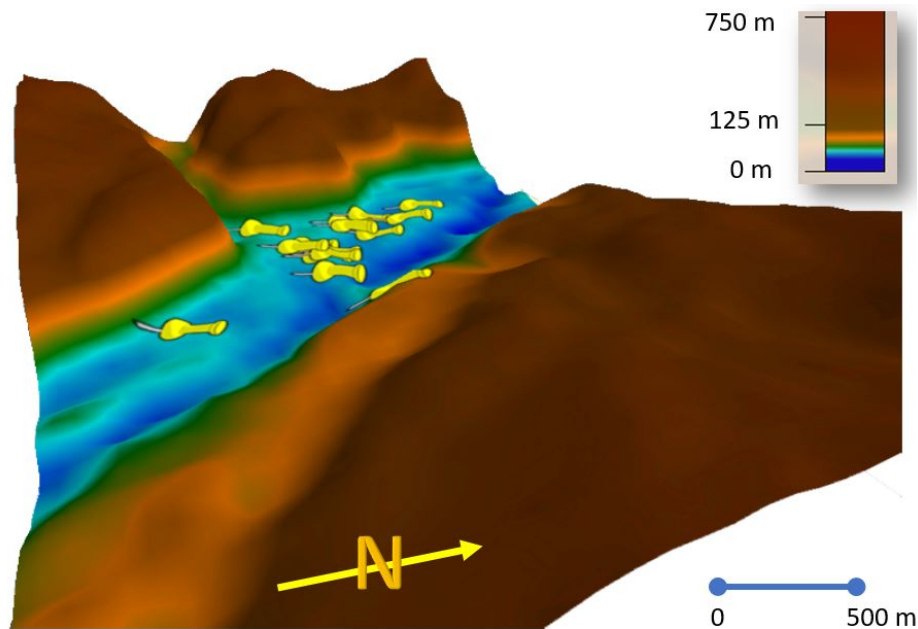
## Methodology

This study area was chosen because of its proximity to the Mohawk River, an area that is prone to frequent floods. The population of Fort Plain was estimated to be 2,248 in 2015 (Census.gov). The target of this study is aimed at businesses and municipal buildings within Fort Plain, NY. The study examines the impact of flooding on businesses and municipal buildings, and if the targeted area would be interested in an application that will enhance the better town's operation and safety upon future flooding. The study was completed via non recorded phone calls using a list of ten questions to approximately forty businesses and municipal buildings. No names or voice was recorded during the surveys. Fifteen successful anonymous phone calls took place using a questionnaire. The questionnaire was designed to focus on whether business and municipal buildings were affected by a previous flood and how useful could a mobile app be during a flood hazard. In order to maintain the anonymity of the interview locations, buffers were created on ArcMap (Fig. 1). Additionally, a digital surface model (DSM) was constructed using the available Shuttle Radar Topographic Mission (SRTM)



**Figure 1:** An aerial view of the study area (from ArcGIS) showing the locations of the interviews with blue buffers.

data for the study area. (Fig. 2). The DSM was constructed in Global Mapper. The points within this flood map were exported as KMZ files from Google Earth. This map illustrates the interview locations within the floodplain.



**Figure 2:** Illustrates a flood map of the interview locations through a digital terrain model (DTM), using Global Mapper and KMZ files. All points are located within the flood plain, as indicated by the blue surface.

The questions asked within the survey are as follows:

1. How many times has your property been affected by a flood event at the past?
2. Do you think that your area could be affected by a future flood event?
3. Did flood warning information reach you in advanced time?
4. How much money were spent at the past 5 years for flood prevention or/and mitigation in your property?
5. During a flood how dependent are you on major roads in your town?
6. There is a need for an application that would automatically notify emergency responders during a flood hazard?
7. I am interested in an app that provides a real time evacuation route during a flood.
8. There is a need for an easily accessible application that says how much and how far the flooding will affect the area?
9. Would you be interested in knowing the exact level of the groundwater on a daily basis around your area using this mobile app?
10. Do you think a mobile application will speed the recovery process after a flood?

The answers ranged from a response of 1-5 categorical answers, where 1 means not interested/low frequency/or /minimal spending, and 5 means strongly interested/high frequency/or/ higher spending (Fig. 3). An extra “Other” category was dismissed as a part of our study to ensure that we only had numerical answers, which would be imported into a workable format of the SPSS software. The calls varied anywhere between 30 seconds to 5 minutes depending on the responders’ willingness to respond. The responses were then imported into a google form and exported to an Excel spreadsheet file. From there they were imported into the SPSS software, where the statistical analysis was completed. Due to the small sample size of 15 responses, the data was re-coded in the statistical software in the following format: all responses that were answered in the first-second-third choice in each individual question (ex. Q1, Q2 ...etc.) were grouped, and in another variable, all the responses that were fourth-fifth choice of the same question were grouped. This was done to ensure that a more accurate comparative analysis could be drawn from the data.

01. How many times has your property been affected by a flood event at the past? \*

1. None, 2. Once, 3. Twice, 4. Three times, 5. More than three times.

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ Other:

**Figure 3:** A screenshot from the questionnaire designed in a Google form to input answers in a database.

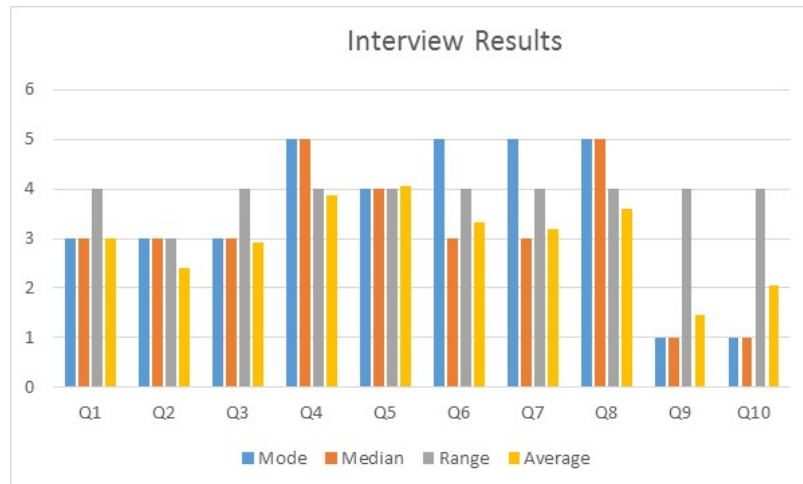
## Results

After the interviews were conducted, the results were analyzed. It was founded that most properties have been affected twice by a flood (Fig. 4) and the interviewees believed that a flood may occur once every five to ten years. In addition, majority of the interviewees were notified a few hours before the flood hazard, they had to pay around \$5,000 to \$10,000 within the last five years for flood prevention/mitigation, and are very dependent on major roads during a flood. These respondents also strongly agreed that there is a need for an application that would notify emergency responders, they were interested in a mobile application that provides evacuation routes, and that there is a need for an application that says how much and how far the flooding will affect the area. Lastly, many people were not interested in knowing the exact level of groundwater on a daily basis and they strongly disagree that a mobile application will speed the recovery process after a flood. The results from the interview are illustrated below (Fig. 4).

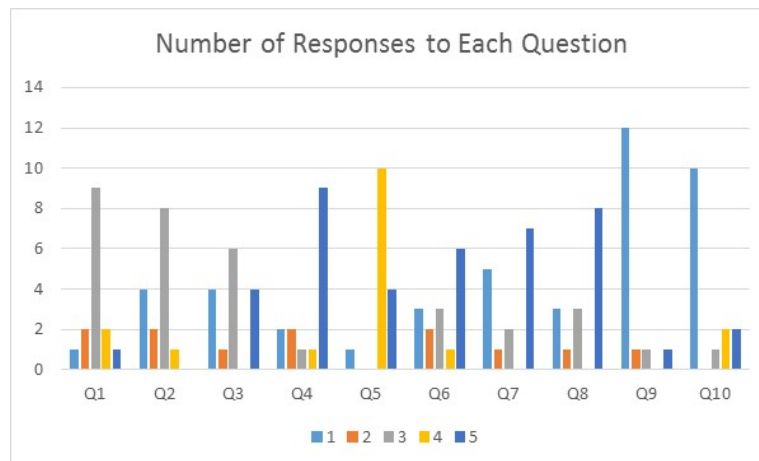
The mode, median, range, and average were also calculated to find the measures of central tendency and variation (Fig. 5). The range is the only measure of variation and illustrated by the grey bar. The mode is the variable that occurs most often and it represented by the blue bar. The median is depicted by the orange bar and is the central variable. Lastly, the average is the sum of the values divided by the number of values, and is represented as the yellow bar. These descriptive statistics are critical to this study because they explain the degree to which all the data values group around or the amount of diffusion.

Subsequently, the data were uploaded into SPSS and a Chi-Square Test was obtained to test the dependency of any of the categorical variable pairs (between questions), and how strongly dependent or independent are two variables (questions) from the utilized questionnaire. In most instances p-values were greater than 0.05 showing the independency between questions, while few questions were dependent with p-values nearby or less than 0.05. Moreover, the questions were tested for their relationship strength using Cramer’s V and Phi number. Cramer’s V is 0 when there is a very weak relationship and have a strong relationship when Cramer’s V is 1. The Chi-Square Test revealed perfect results between questions one and five; the p-value was 0.005 and Cramer’s V was 1.000 (Fig. 6).





**Figure 4:** Illustrates the number of responses (frequency) (1-5) to each question.



**Figure 5:** Mode, median, range, and average for each of the interview question results.

|                              | Value               | df | Asymptotic Significance (2-sided) |
|------------------------------|---------------------|----|-----------------------------------|
| Pearson Chi-Square           | 15.000 <sup>a</sup> | 4  | .005                              |
| Likelihood Ratio             | 7.348               | 4  | .119                              |
| Linear-by-Linear Association | 5.000               | 1  | .025                              |
| N of Valid Cases             | 15                  |    |                                   |

a. 9 cells (90.0%) have expected count less than 5. The minimum expected count is .07.

|                    |            | Value | Approximate Significance |
|--------------------|------------|-------|--------------------------|
| Nominal by Nominal | Phi        | 1.000 | .005                     |
|                    | Cramer's V | 1.000 | .005                     |
| N of Valid Cases   |            | 15    |                          |

**Figure 6:** Chi-Square Test results, obtained from SPSS, which demonstrate Cramer's V and P-value.

## **Discussion**

The correlation between questions one and five illustrates that the higher occurrence of floods results in a high dependency on major roads. The greater the incidence of flooding hazards, the interviewees expressed how important major roads are for escape routes. Many respondents expressed that the major roads were the only method of travel. Throughout the interviews, various respondents reiterated that the Federal Emergency Management Agency (FEMA) aided business owners in repairs and preparations for future storm events. Most of the business owners (interviewees) received money from this agency. It was also founded that many of the respondents were confused by the term groundwater and its relevance to this study. This implied a lack of fundamental knowledge around the groundwater flood formation. Fort Plain workers also seemed hesitant toward the notion of a mobile application speeding the recovery process. One respondent stated that the government would be more beneficial toward recovery than a mobile application. The low interest in having a mobile application to facilitate a recovery after a flood hazard illustrates the lack of understanding of how dependent we are on roads and the local infrastructure. This includes the operation of the police and fire departments, mail carriers, hospital operations, and ambulances upon a flood hazard.

## **Conclusion**

Many businesses lie within the floodplain and are at risk for future flooding hazards. The last major flooding hazard within this area was in June 2013 but remediation is still underway. Although this area is frequently affected by floods, many municipal workers and business owners were not interested in participating for the interviews or knowing important features such as the groundwater level. The successful interviews, however, demonstrate that there is a need for a mobile application that would notify emergency responders, provide evacuation routes, and alert workers how much and how far the flooding will affect or affects Fort Plain during a flood hazard. This mobile application will be pertinent to this area, and will create emergency preparedness and mitigation. Additionally, the application will inform the workers where evacuation shelters are or how to approach them through safe and non-flooded routes. Understanding the flood extent in an area during or after a flood hazard such as in Fort Plain will enhance community resilience against future floods.

## **Acknowledgement**

We thank the Institutional Review Board (IRB) of Hofstra University's approval for the use of humans as research objects and for assisting us in ensuring that our survey met the criteria.

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Poster Presentation